FRI Natural language processing course 2026
Initial report

University *of Ljubljana*
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Natural language processing - Submission 1: Project selection and simple corpus analysis

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Abstract

General-purpose LLMs often lack knowledge of specific legal systems and are rarely tailored to smaller jurisdictions. In this project, we will develop a conversational AI assistant for Slovenian law using retrieval-augmented generation (RAG) and prompt engineering, ensuring that answers are grounded in actual legislation. The system is designed to support practical tasks such as contract preparation and understanding legal provisions. The report describes the dataset, system design, and evaluation, providing insights into building domain-adapted AI for legal applications.

Introduction

General purpose large language models (LLMs) struggle with domain specific tasks such as legal advising, as they lack access to authoritative and jurisdiction specific sources. Even if said models exist they are usually only usable for certain larger countries. This project develops a conversational AI assistant tailored to Slovenian law, leveraging the legal corpus prepared within the LLM4DH initiative. Using retrieval augmented generation (RAG) and prompt engineering, the assistant grounds its answers in actual legislation and supports realistic use cases such as contract preparation or statutory guidance. The report details the data curation, system architecture and evaluation of the assistant, offering insights into building reliable, domain adapted AI systems for legal applications.

Existing solutions and related work

Natural language processing in the legal domain has developed from basic information retrieval and classification toward more advanced systems for question answering, summarization, and general legal assistance. However, legal texts are still challenging to work with. They are often long, highly structured, and full of specialized terminology, and their meaning depends heavily on jurisdiction and context. Because of this, methods that work well in general NLP cannot be directly applied to legal tasks without careful adaptation.

Practical legal-support systems provide a useful point of comparison, two Slovenian examples being MojMaliPravnik and Pravko. MojMaliPravnik allows users to ask legal questions and browse existing answers, while Pravko is an AI-supported assistant that helps users navigate legal and administrative topics such as legislation, labour law, real estate, inheritance, and taxes [1, 2]. These systems show that there is real demand for tools that make legal information more accessible to non-experts. At the same time, they highlight an important requirement: answers should be clear, reliable, and grounded in authoritative sources.

For LLM-based legal assistants, one of the main challenges is hallucination. In the legal domain, incorrect statements or fabricated citations can have serious consequences. This makes retrieval-based approaches particularly important. Retrieval-Augmented Generation (RAG) combines a language model with a document retrieval step, so that answers are generated based on relevant source texts rather than only from the model's internal knowledge [3]. In legal applications, this is especially useful because it allows the system to provide supporting sources and remain closer to official legal texts.

Overall, the related work suggests that a useful legal assistant should be both grounded and specialized. Legal tasks require more than fluent text generation, and recent work highlights the importance of source-based answering and safeguards against hallucinations [4, 5, 6, 7]. Based on this, our project focuses on building a domain-specific assistant for Slovenian legislation that retrieves relevant legal texts and provides grounded answers for use cases such as legal question answering or contract-preparation support.

Proposed dataset and data

This project utilizes the COLESLAW 1.0 corpus [8], a large collection of Slovenian legal texts developed within the LLM4DH initiative and made available through CLARIN.SI. The corpus contains over 500,000 documents and approximately 770 million words and is provided in JSONL format with cleaned text and metadata. The dataset includes sources such as ”PISRS (legislation)”, ”Uradni list”, ”USRS” and ”Sodna praksa”.

Since our goal is to build a domain specific assistant for Slovenian legislation, we treat the corpus as the main source of truth, but focus primarily on legislation oriented data (especially PISRS) in the initial version. The JSONL structure allows us to preserve metadata and trace answers back to original legal sources. In the next steps, we will preprocess the data by selecting the relevant subset and splitting documents into smaller units (e.g. articles or sections) suitable for retrieval. The corpus combines different types of legal documents, for this reason, we begin with a restricted subset and consider using the rest in later stages.

Initial ideas

Our goal is to build a legal question-answering assistant for Slovenian legislation that provides answers grounded in actual legal texts. To do this, we will use a Retrieval-Augmented Generation (RAG) approach.

We will begin with a simple baseline system based on a legislation-focused subset of the COLESLAW corpus, mainly PISRS. The documents will be preprocessed and split into smaller units, such as articles or sections, and then indexed using embeddings to enable semantic search. When a user asks a question, the system will first retrieve the most relevant legal passages and then generate an answer based only on this retrieved content. We will carefully design prompts so that responses are clear, accurate, and supported by the original sources. If the system cannot find enough relevant information, it will return a fallback response instead of generating potentially incorrect answers (to prevent hallucination).

We will evaluate the system by looking at both how well it retrieves relevant documents and how accurate the final answers are. We will also compare it with a non-grounded LLM baseline to better understand the benefits of using retrieval. As a possible extension, we may experiment with improvements such as lightweight domain adaptation methods (PEFT, LoRA), but our main focus is on building a reliable baseline system.

References

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